Performance and Robustness Benchmarking Across Battery State-of-Charge Estimator Classes for Embedded Applications

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Abstract

Embedded battery-state estimators require validation under disturbed inputs and the inference semantics used on resource-constrained hardware. This study presents a causal, cell-disjoint benchmark of four State-of-Charge (SOC) approaches: fixed-capacity Coulomb counting as a direct-measurement model (DM), Coulomb counting with data-driven State-of-Health adaptation as a hybrid direct-measurement model (HDM), a second-order resistor-capacitor observer as a hybrid equivalent-circuit model (HECM), and a gated recurrent unit structurally pruned from 96 to 67 hidden units as the data-driven model (DD). Public lithium iron phosphate ageing data cover multiple operating and health conditions. Robustness evaluation addresses nominal accuracy, corrupted or unavailable measurements, and recovery from initialization errors. Hardware evaluation on an STM32H753 measures numerical equivalence, latency, flash, and peak runtime memory. For DD, exact rolling-window recomputation, continuous state forwarding, and periodic state reset are compared. Nominal cell-macro mean absolute errors are 0.0690, 0.0412, 0.0337, and 0.0258 for DM, HDM, HECM, and DD. DD yields the lowest nominal error and leads several disturbance scenarios, whereas HECM shows the strongest recovery. All implementations reproduce

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their software references, fit available memory, and meet the one-second sampling deadline. Compared with rolling-window recomputation, continuous-state DD reduces median latency from approximately 724 ms to 0.425 ms and peak memory from 67.3 KiB to 4.1 KiB while retaining similar nominal error. Periodic resets retain these resource benefits but cause pronounced transients. The results demonstrate that estimator choice, model compression, and recurrent-state handling must be evaluated jointly with accuracy, robustness, recovery, and embedded cost.

Keywords: State-of-charge estimation, Battery management system, Recurrent neural network, Robustness benchmarking, Embedded artificial intelligence, Model compression

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Nomenclature

Acronyms

BMS	Battery management system
SOC	State of charge
SOH	State of health
DM	Direct-measurement model
HDM	Hybrid direct-measurement model
HECM	Hybrid equivalent-circuit model
DD	Data-driven neural model
ECM	Equivalent circuit model
EKF	Extended Kalman filter
MAE	Mean absolute error
RMSE	Root-mean-square error
ADC	Analog-to-digital converter
CV	Constant-voltage phase
OCV	Open-circuit voltage

GRU	Gated recurrent unit
LSTM	Long short-term memory
LFP	Lithium iron phosphate

Symbols

t	Time
$I(t)$	Current signal at time t
$U(t)$	Voltage signal at time t
$T(t)$	Temperature signal at time t
$\widehat{\mathrm{SOC}}(t)$	Estimated state of charge at time t
$\widehat{\mathrm{SOH}}(t)$	Estimated state of health at time t
$\mathrm{SOC}_{\mathrm{ref}}(t)$	Dataset SOC ground truth at time t
$\mathrm{SOH}_{\mathrm{ref}}(t)$	Reference SOH target at time t
g_I	Multiplicative current-gain error
ΔI	Additive current offset
ΔU	Additive voltage offset
ΔT	Additive temperature offset
σ_I	Standard deviation of injected current noise
σ_U	Standard deviation of injected voltage noise
σ_T	Standard deviation of injected temperature noise
Q_c	Online cumulative charge state used by the SOC models
C_{nom}	Nominal capacity
C_{eff}	Effective capacity used inside the estimator
$U_{\max}$	Upper voltage threshold used for full-charge detection
U_{tol}	Voltage tolerance used in the full-charge condition
U_1, U_2	Polarization voltages of the first and second RC branch
ΔMAE	Error increase relative to baseline

1. Introduction

1.1. Motivation and practical relevance

Reliable battery state estimation is a core BMS function because SOC and SOH determine safety margins, usable energy, operating limits, charge control, and degradation-aware system management in both mobile and stationary lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not defined by clean-condition accuracy alone, because real BMS operation is exposed to sensor gain and offset errors, additive noise, initialization uncertainty after restart, missing samples, irregular sampling, communication interruptions, and transient signal spikes caused by measurement-chain limitations or disturbances in the surrounding system [4, 5, 6]. The relevant engineering question is therefore not only whether an estimator achieves low MAE under nominal measurements, but also whether it converges back to a reliable state after disturbed initialization and whether it remains stable under physically plausible measurement corruption.

As summarized conceptually in Figure 1, the present benchmark treats deployment-facing estimator quality as a three-dimensional problem consisting of nominal accuracy, recovery after state inconsistency, and robustness against disturbed measurements and signal-integrity faults, embedded within broader constraints on memory and execution time. The central question is how four structurally different SOC-estimation classes, represented by the implementations studied here, respond to common failure mechanisms and microcontroller constraints, and whether their nominal ranking remains informative under those conditions. This distinction is essential because one implementation can be accurate under clean measurements yet difficult to re-anchor, sensitive to a specific input fault, or too costly under its trained inference semantics.

1.2. State of the art in battery state estimation

Battery-state estimators are commonly grouped into four broad families [7, 1, 2, 3]. These are direct or integration-based methods, model-based methods using

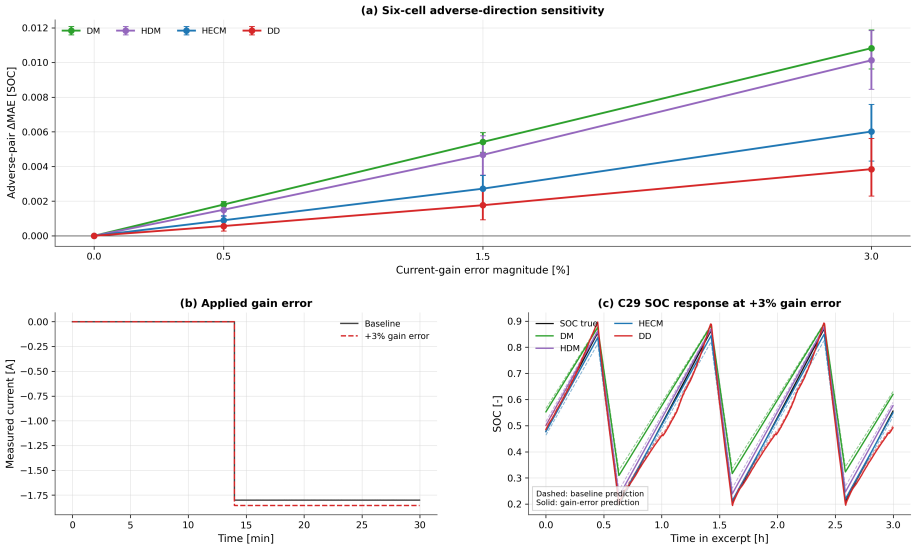

Figure 5: Current-gain sensitivity. (a) Six-cell adverse-direction ΔMAE for each matched $\pm g_I$ pair with hierarchical 95% confidence intervals. (b) Representative current before and after the +3% gain error. (c) SOC trajectories from the representative high-load test cell over three charge–discharge intervals. The test-set load assignment is documented in Figure A.21. The adverse-pair summary reports the larger penalty from each matched sign pair.

transferred charge. HECM is exposed through the same current-driven state propagation, but its voltage-residual correction can counter part of the resulting state deviation. DD receives current together with voltage, temperature, cumulative charge, derivatives, and temporal context, which is consistent with its smaller observed dependence on one systematically scaled channel. The complete signed sweep remains available for interpreting direction-specific behavior.

The continuous high-load test-set analysis in Figure 6 complements the window-based benchmark and separates three mechanisms that independent 24-h windows cannot resolve. The gain-error contribution varies over life because its signed Coulomb-counting error interacts with charge direction and operating state. Between full charges, the penalty often grows within an interval, but full-charge anchoring can partially reduce or restructure it. The process is therefore neither an unrestricted monotonic drift nor a complete reset. Across

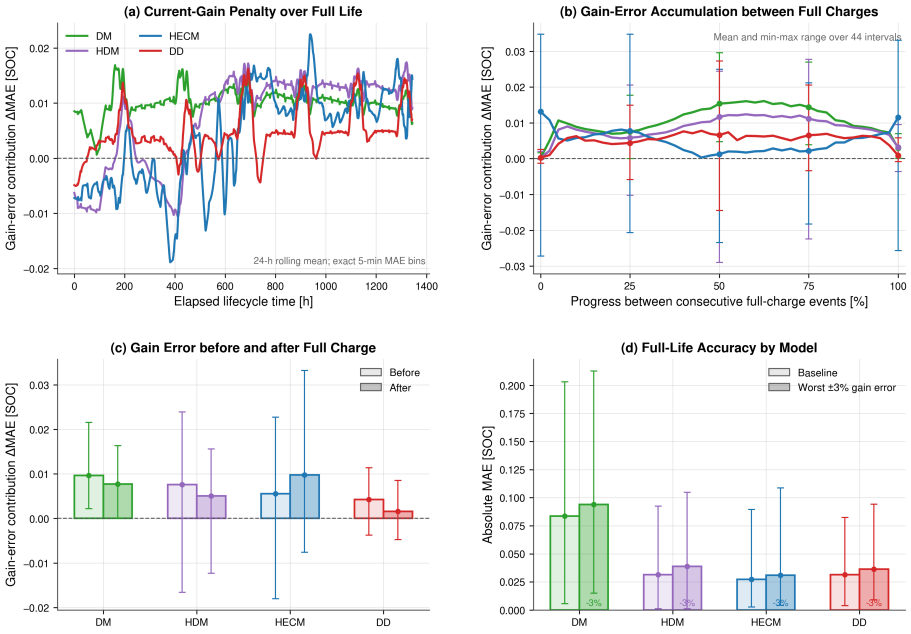

Figure 6: Current-gain-error diagnostic over the continuous high-load holdout trajectory. (a) Rolling 24-h gain-error contribution over elapsed life. (b) Mean and range over normalized intervals between consecutive full-charge events. (c) Penalty immediately before and after full-charge events. (d) Full-life baseline and adverse-gain MAE. Full-charge events reduce or restructure accumulated error but do not erase the lifetime penalty uniformly.

this single full-life replay, the adverse 3% gain error increases MAE by 0.0102 for DM, 0.0072 for HDM, 0.0037 for HECM, and 0.0048 for DD. HECM therefore has the smallest full-life gain-error contribution in this trajectory, while DM and HDM remain most exposed. This mechanism-focused diagnostic is not a six-cell lifecycle ranking.

4.3. Additive sensor offsets

The signed current-offset experiment shifts the measured current by ± 0.05 A and selects the larger paired ΔMAE within each cell before equal-weight aggregation. DD has the smallest adverse cell-macro penalty at 0.0401 (95% CI

0.0226–0.0556), followed by HECM at 0.1763 (0.1329–0.2142), DM at 0.2467 (0.2253–0.2693), and HDM at 0.2727 (0.2546–0.2911). The adverse sign differs across models and cells, which confirms that a single favorable sign would not represent offset sensitivity reliably. The large response of the integration-based implementations follows from accumulating a persistent zero-current error throughout the evaluation window, whereas the recurrent DD implementation limits but does not eliminate this systematic shift. Among the evaluated measurement disturbances, additive current offset produces the largest global cell-macro penalty for all four representatives. The complete comparison with the remaining disturbance scenarios is shown in Figure 15.

The fixed positive voltage and temperature offsets produce a different model pattern. A $+3\,^{\circ}\mathrm{C}$ temperature offset increases DD MAE by 0.0074 (95% CI 0.0027–0.0125), while the corresponding changes for DM, HDM, and HECM are 0.0000, 0.0002, and 0.0002. A $+0.02\,\mathrm{V}$ voltage offset increases DD by 0.0038 and HDM by 0.0005, leaves DM effectively unchanged, and changes HECM by -0.0040 in these finite windows. This pattern follows the model-specific use of the shifted channels. DM does not use temperature and uses voltage primarily for physical anchoring rather than continuous propagation. HDM receives both channels indirectly through the slowly updated SOH service. HECM responds to voltage through its observer residual, where the fixed offset can partly compensate an existing trajectory error over a finite window. DD uses absolute voltage and temperature as direct learned inputs and therefore transmits both systematic shifts into the SOC prediction. The signed HECM result must consequently be interpreted as compensation relative to its baseline rather than an improvement in the physical model. The additive current, voltage, and temperature offsets form one offset family in the robustness synthesis, with the signed current case represented by its adverse paired result.

4.4. Noise sensitivity

In contrast to the persistent gain and offset cases, the sensor-noise experiments represent random zero-mean fluctuations. They produce smaller global

penalties than the systematic current errors, with a response that depends on the disturbed channel and estimator structure. At $\sigma_I = 0.10\,\mathrm{A}$, HECM has the largest mean macro penalty ($\Delta\mathrm{MAE} = 0.0038$), although its interval spans zero. DM changes by -0.0009, while HDM and DD remain near zero. In DM and HDM, every disturbed current sample enters the cumulative charge state Q_c, but positive and negative noise increments partly cancel during integration. The integration therefore smooths rapid zero-mean fluctuations in the resulting SOC trajectory even though no individual sample is ignored. HECM processes disturbed current directly in both state propagation and the voltage-prediction path, which provides a plausible transmission mechanism for its larger mean response. DD also receives the disturbed raw current directly and, unlike DM and HDM, additionally receives $\mathrm{d}I/\mathrm{d}t$ at every recurrent step. These direct channels explain the visibly larger short-term DD output variation in Figure 7(d). The rolling multichannel context distributes this response over the input sequence. At the tested level, the local transmission does not develop into a persistent global bias, so DD remains near zero in the window-wide ΔMAE. These explanations remain conditional on the reported uncertainty because the HECM interval includes zero. Voltage noise ($\sigma_U = 0.01\,\mathrm{V}$) produces a 0.0014 penalty for DM and 0.0013 for HDM, while HECM and DD remain below 0.0003. The small DM and HDM voltage response can arise through voltage-dependent full-charge anchoring rather than their continuous current-integration update. Temperature noise is neutral for DM and small for the SOH-aware models because temperature enters those pipelines through learned or parameter-dependent pathways rather than the direct charge update. Figure 7 shows the local transmission of current noise, while Figure 8 reports repeated six-cell effects with hierarchical uncertainty.

Taken together, the repeated six-cell results show that zero-mean sensor noise causes only small changes in window-wide accuracy for most model–channel combinations. The local current-noise analysis further demonstrates that window-wide ΔMAE should be interpreted together with short-term output transmission because temporal fluctuations may be only weakly represented

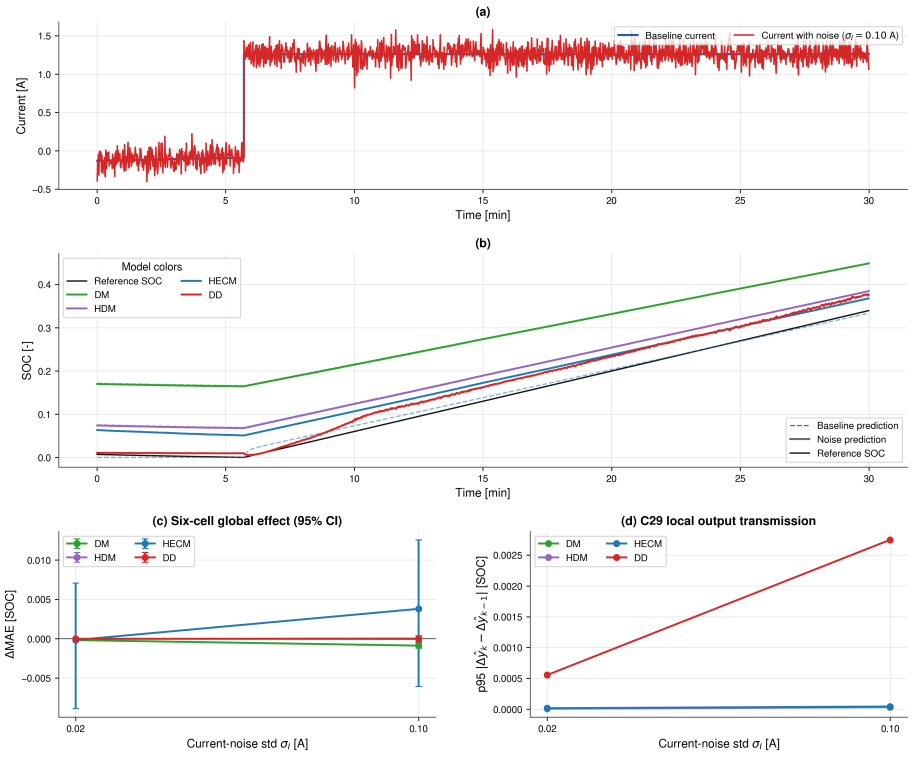

Figure 7: Noise sensitivity under additive current noise. (a) Baseline and noisy current in a reset-free 30 min example from the high-load test cell. (b) Matched baseline and noisy SOC trajectories in the same window. (c) Cell-macro ΔMAE with hierarchical 95% confidence intervals across the holdout test set. (d) Local transmission in the same example, measured as the 95th percentile of the change in noise-induced output deviation between consecutive samples. DD shows the largest local transmission because raw current and its derivative enter the recurrent input directly, whereas the integration in DM and HDM smooths zero-mean sample-level fluctuations. The local panels illustrate the mechanism, while panel (c) provides the test-set comparison.

by an aggregate error metric.

4.5. Initialization error and recovery

Initialization performance uses the paired criterion defined above. The protocol applies the same −0.10 SOC-equivalent mismatch through the deployment-accessible state of each estimator. It offsets the explicit Coulomb-counting state

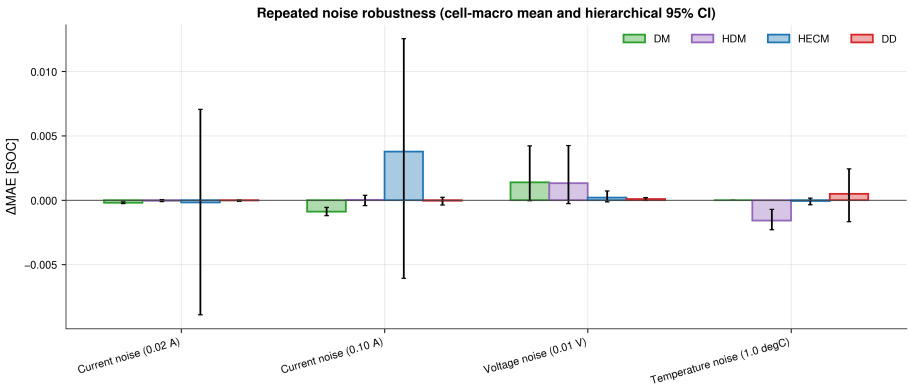

Figure 8: Repeated noise robustness across the six-cell holdout test set. Bars show cell-macro ΔMAE and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel–model combinations remain close to baseline relative to the between-cell uncertainty.

in DM and HDM and the EKF SOC state in HECM. For DD, it applies the equivalent cumulative-charge shift $Q_c - 0.10C_{\mathrm{nom}}$. Figure 9 reports the resulting paired trajectory differences from the common evaluation start, while recovery time remains referenced to the intervention. HECM provides the fastest persistent recovery, with a cell-macro mean of 1.20 h, followed by DD at 2.60 h. DD enters the recovery band temporarily in several runs but has a higher relapse fraction than HECM (0.78 compared with 0.17). DM and HDM generally do not recover persistently within the observation horizon, and both have a persistent-censor fraction of 0.89. HECM also has the smallest excess-error area at 0.078 SOC h, compared with 0.116 SOC h for DD. The recovery pattern follows the available correction mechanisms. DM and HDM propagate the imposed state offset until a physical charge anchor is reached because neither method has continuous output feedback. HECM uses the voltage residual to correct its explicit SOC state and can therefore re-anchor persistently. DD does not propagate SOC as an isolated state. Its multichannel rolling input allows the prediction to move away from the disturbed cumulative-charge feature, although the interaction of that feature with the changing recurrent context explains why

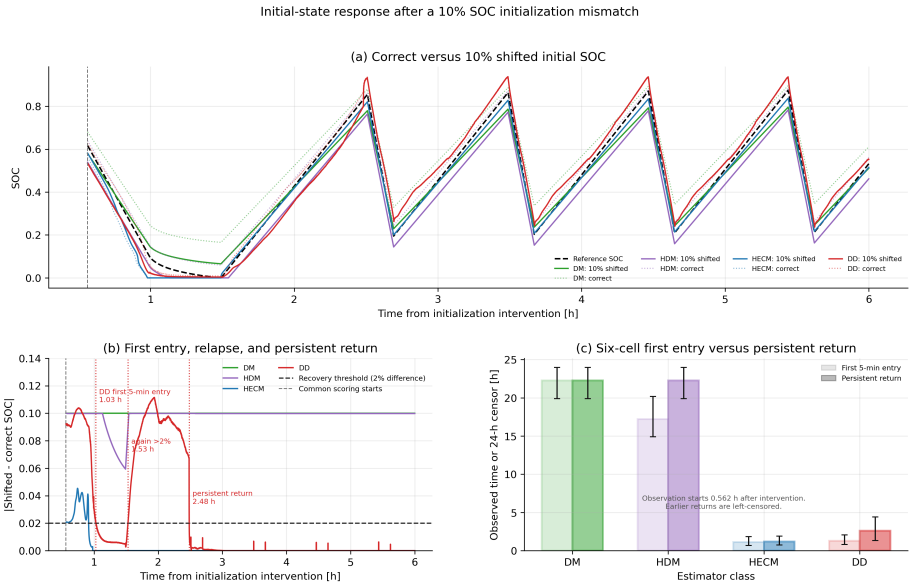

Figure 9: Response to a 10% SOC-equivalent initialization mismatch. (a) Correctly and incorrectly initialized trajectories for a representative mid-life pair from the high-load holdout trajectory. (b) Absolute paired differences over the first 6 h. The DD trajectory enters the 2% band for 5 min, leaves it again, and only later returns persistently. The vertical gray marker denotes the common evaluation start after all estimators provide valid output. (c) Observed first qualifying entry and persistent recovery-or-censor time across the holdout test set with hierarchical 95% confidence intervals. Endpoints already satisfied at the common start are left-censored.

entry into the recovery band can be followed by relapse. Together, the recovery time, relapse behavior, censoring, and excess-error area identify HECM as the strongest recovery implementation under this protocol.

4.6. Missing samples and signal integrity

Figure 10 compares the six-cell effects of periodic and random sample loss with those of timing jitter. Sample loss is substantially more consequential than timing jitter. Periodic freezing of every 50th sample increases MAE by 0.0072, 0.0065, 0.0028, and 0.0011 for DM, HDM, HECM, and DD, respectively. Random 2% loss produces corresponding increases of 0.0080, 0.0081, 0.0039, and 0.0026. By contrast, even the $\pm 0.9\,\mathrm{s}$ jitter case remains within approximately

6×10^{-4} SOC of baseline for every model, with confidence intervals generally spanning zero. DD therefore has the smallest sample-loss penalty, while DM and HDM retain their structural dependence on uninterrupted charge integration. A missing or frozen current sample removes or distorts charge information that an integration-based estimator cannot reconstruct retrospectively. HECM can reduce part of this error through voltage feedback, while DD can use the surrounding multichannel sequence rather than relying on the affected sample alone. Timing jitter preserves the measurements themselves, and each implementation reconstructs its time-dependent quantities from the disturbed sampling intervals. This explains why timing perturbation is less consequential than information loss under the tested protocol.

The one-hour burst dropout creates a distinct availability problem. Against the duration-matched 48-h baseline, its macro ΔMAE is largest for HECM (0.0239, 95% CI 0.0155–0.0366), followed by HDM (0.0209, 0.0098–0.0313) and DM (0.0050, −0.0029–0.0138). DD changes by −0.0093 (−0.0288–0.0099), reflecting finite-window compensation between its baseline and frozen-input trajectories. The DD interval includes zero, so the negative mean does not demonstrate an accuracy improvement. It shows that no persistent global degradation is resolved under this protocol. The interval is placed where the absolute unobserved net charge is largest among starts that retain the required pre- and post-gap context. Figure 11 compares the disturbed output of the representative high-load test cell with its duration-matched no-dropout trajectory and reports the global holdout-test effect. DM and HDM retain the corrupted charge balance until a physical anchor because the missing interval cannot be reconstructed by integration alone. HECM receives measurements again after the gap, but the flat LFP voltage relation limits how strongly its observer can infer the missing charge and correct the accumulated state deviation. DD progressively removes frozen raw and derivative samples from its finite rolling window after measurements resume. Its voltage, temperature, current, and temporal context can then reduce dependence on the corrupted cumulative-charge feature. This mechanism is consistent with its smaller persistent effect, while the

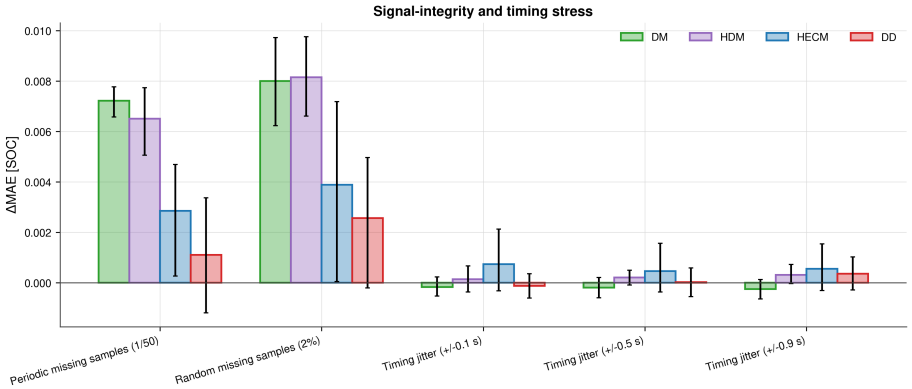

Figure 10: Six-cell signal-integrity and timing stress. Bars show cell-macro ΔMAE and hierarchical 95% confidence intervals for periodic and random sample loss and three timing-jitter levels. Sample loss clearly dominates timing jitter under the tested protocol.

finite-window confidence interval prevents a stronger claim. Dropout contributes to the robustness dimension, while the recovery dimension uses the canonical paired initialization experiment.

4.7. Spike sensitivity

Single voltage spikes demonstrate why local and global metrics must be reported together. Full-window ΔMAE remains below 4×10^{-4} SOC for every estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis, however, identifies a reproducible DD transient: its mean spike-sample penalty across cells is approximately 7.7×10^{-3} SOC, compared with near-zero direct penalties for the other representatives. Figures 12 and 13 further show that the DD response varies across cells and SOH states and then decays after the event. DM and HDM do not use voltage as a continuous SOC-correction signal, so an isolated spike away from the full-charge condition has little direct effect. HECM processes voltage through its observer residual, which filters a single inconsistent measurement through the model and covariance weighting. DD receives the impulse simultaneously through absolute voltage and $\mathrm{d}U/\mathrm{d}t$, and the disturbed values enter its rolling recurrent context. This provides a plausible mechanism for the larger but short-lived DD response. A channel-removal ablation would be

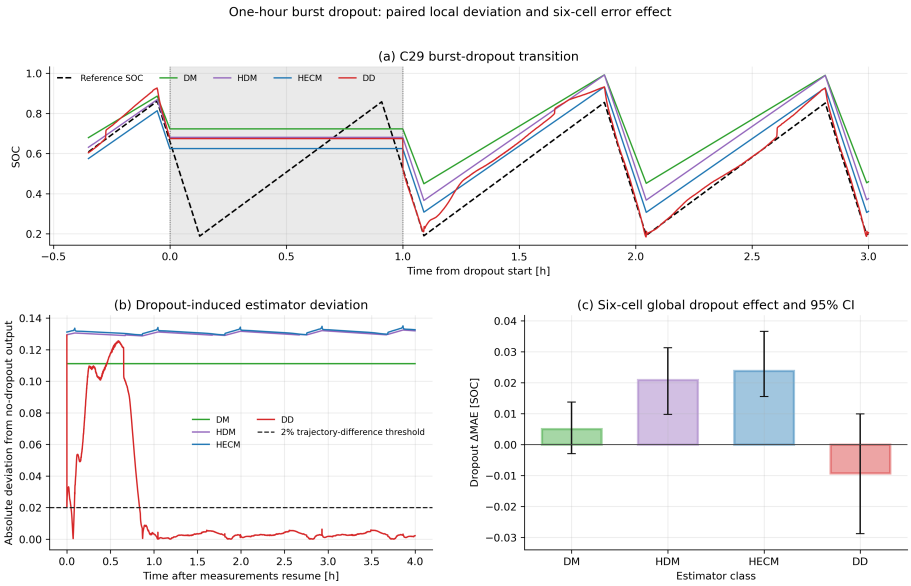

Figure 11: One-hour burst-dropout transition. (a) Estimator outputs around the shaded missing-input interval. (b) Absolute deviation from the paired duration-matched no-dropout output after measurements resume, with the 2% trajectory-difference line shown for context. (c) Six-cell global ΔMAE with hierarchical 95% confidence intervals.

required to separate the individual contributions of the absolute and derivative inputs.

4.8. ADC quantization

Figure 14 compares representative raw and quantized current and voltage signals and summarizes the resulting six-cell ΔMAE. The tested ADC steps ($\Delta I = 0.01\,\mathrm{A}$, $\Delta U = 0.005\,\mathrm{V}$, and $\Delta T = 0.5\,^{\circ}\mathrm{C}$) preserve the local signal shape but affect the tested representatives differently. DM and HDM increase by 0.0038 and 0.0035 MAE, respectively. HECM changes by -0.0018 and DD by $+0.0002$. All four confidence intervals overlap zero. The small DM and HDM response is consistent with repeated current-rounding errors entering their cumulative charge states. HECM combines the quantized current with voltage-residual correction, while DD embeds all quantized channels in a multichannel temporal context. Their signed changes remain within the uncertainty range

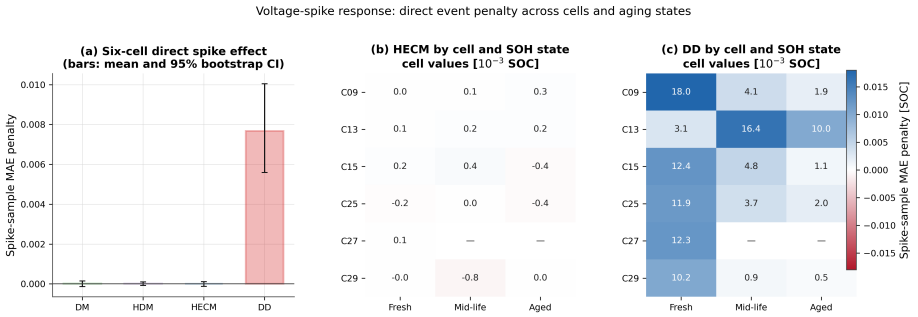

Figure 12: Voltage-spike sensitivity across six cells. (a) Mean event-sample MAE penalty with hierarchical 95% confidence intervals. (b,c) Cell- and SOH-state-resolved penalties for HECM and DD, respectively. Absent cell/state combinations are left blank. The event-aligned metric exposes the local DD response that is obscured in full-window ΔMAE.

and must not be interpreted as physical improvement or exact cancellation. At the selected coarse resolution, quantization produces smaller penalties than current gain error, sample loss, and burst dropout, while the integration-based representatives retain a measurable response.

4.9. Diagnostic comparison with ground-truth SOH

The primary comparison supplies the same causally estimated LSTM-SOH trace to HDM, HECM, and DD. This estimated trace represents the health information available to the complete estimator pipelines during online operation. The diagnostic comparison replaces it with the dataset reference SOH, which is the ground-truth SOH derived from capacity checkups and interpolated between the measured anchors. Its purpose is to identify whether observed SOC errors originate predominantly within the SOC branch or are inherited through the estimated health input.

Reference SOH reduces baseline MAE by 0.0370 for HDM and 0.0166 for HECM, while DD MAE increases by 0.0049 on average. The pronounced HDM improvement is consistent with its direct use of SOH to scale effective capacity. HECM also benefits because SOH enters both capacity scaling and lookup-based parametrization. DD instead uses SOH as one learned feature among

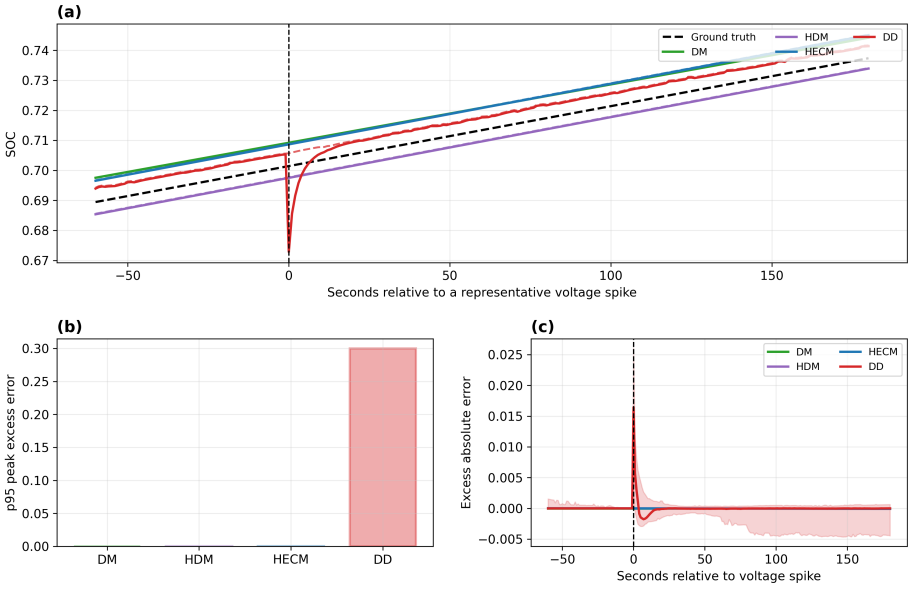

Figure 13: Event-aligned voltage-spike response. (a) Representative SOC trajectories around the injected spike, with the dashed black line showing dataset ground-truth SOC. (b) Model-wise 95th percentile peak excess error. (c) Mean excess absolute error aligned across repeated spikes, with the DD uncertainty envelope. The local response decays rapidly and is therefore weakly represented by full-window MAE.

several interacting inputs. Replacing this single feature with its ground truth therefore does not impose a monotonic relationship between SOH accuracy and the resulting SOC error.

The same model-specific direction persists for most noise, sample-loss, and dropout cases. At 3% current gain error, reference SOH improves HDM and HECM and changes DD by +0.0091. For temperature noise, the reference-minus-LSTM difference changes from its baseline value by 0.0016 for HDM and less than 10^{-4} for HECM and DD. The incremental temperature-noise response is therefore largely independent of the SOH estimator. These findings show that SOH-input quality explains part of the baseline differences, especially for HDM and HECM, but does not account for every disturbance-specific response. The LSTM-SOH runs consequently remain the primary deployment-representative

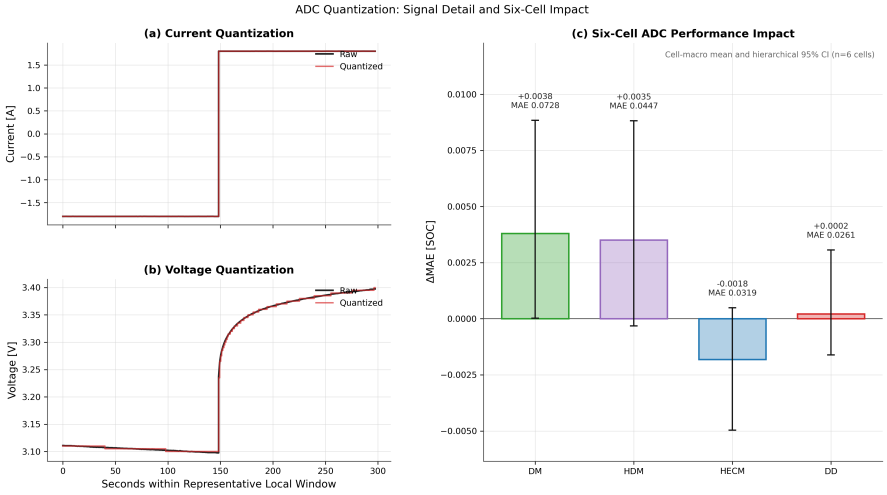

Figure 14: ADC quantization. (a,b) Representative raw and quantized current and voltage. (c) Six-cell ΔMAE with hierarchical 95% confidence intervals and resulting quantized-signal MAE annotations.

results, while the ground-truth SOH runs provide the diagnostic attribution.

4.10. HECM lookup-table sensitivity

Within the tested local perturbation range, lookup-table uncertainty is not the dominant source of the observed HECM measurement-disturbance penalties. Across all 22 measurement and signal-integrity subcases, the largest absolute lookup–disturbance interaction is 0.006116 MAE. It occurs within the additive current-offset test, but its cell-bootstrap interval includes zero. This interaction is small relative to the HECM current-offset penalty of 0.1763 MAE obtained with the nominal lookup. The large offset response is therefore attributable primarily to the systematic measurement error and not to the tested variations of the parameter surfaces.

The lookup tables nevertheless remain relevant to baseline calibration and recovery. Five of the six lookup perturbations retain mean persistent recovery-or-censor times between 1.18 and 1.41 h, close to the nominal value of 1.20 h. Under the remaining resistance perturbation, one low-load holdout cell stays

outside the recovery band for the complete observation horizon and raises the equal-weight mean to 5.10 h. The analysis therefore answers the primary disturbance-attribution question negatively while identifying recovery as a separate calibration-sensitive boundary. Figure B.26 and Table B.5 provide the complete Appendix results.

4.11. Robustness synthesis across representatives

Figure 15 condenses 18 measurement-corruption and signal-integrity rows and shows how degraded-input behavior changes the nominal-accuracy picture. The additive current offset creates the largest global penalties for DM, HDM, and HECM, while DD remains less affected. Current-gain sensitivity increases with magnitude for every representative in the matched signed sweep. Periodic and random sample loss particularly affect DM and HDM, and the one-hour dropout produces the largest non-offset point penalty for HECM. Timing jitter remains nearly neutral. Voltage spikes reveal a distinct DD transient through event alignment despite their small full-window MAE. Figure 16 complements this scenario-level comparison by combining relative accuracy, family-balanced robustness, and recovery and by showing how alternative priority profiles change the overall assessment. Together, both figures provide the evidence required for deployment-oriented estimator selection.

Under the family-balanced definition, the robustness scores are 0.452 for DM, 0.351 for HDM, 0.360 for HECM, and 0.967 for DD. Equal weighting per evaluated scenario yields 0.549, 0.476, 0.552, and 0.793, while the highest-level-plus-offset variant yields 0.573, 0.469, 0.562, and 0.693. DD leads all three aggregations. The changing score distances and the changing order of DM and HECM demonstrate the influence of the aggregation rule, while the scenario-level results identify the disturbance mechanisms behind these relative summaries.

4.12. Embedded performance and recurrent execution modes

The STM32 experiment first verifies numerical equivalence before comparing speed or memory. Across the six nominal cell sequences, mean hardware–software MAE remains below 4×10^{-4} percentage points for every SOC core

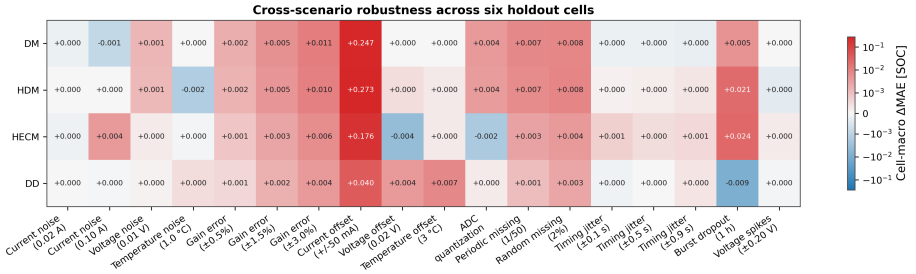

Figure 15: Cell-macro ΔMAE across 18 measurement-corruption and signal-integrity rows on the six-cell holdout test set. Current-gain and additive current-offset rows use adverse directions from matched signed sweeps. A symmetric logarithmic color scale retains visual resolution for smaller effects despite the larger current-offset penalties. Positive values denote degradation relative to each model's paired baseline, while small negative values arise from finite-window signed compensation. The initialization-mismatch response is evaluated separately in Figure 9.

(Figure 17). These differences are orders of magnitude below the estimator-to-dataset errors and show that the float32 C implementations reproduce their software references sufficiently closely for deployment measurements.

The isolated DM and HDM SOC updates both have sub-microsecond median latency (0.171 and 0.165 μs), HECM requires 23.9 μs, and continuous-state DD requires 424.9 μs (Figure 18). The latency order follows the computation performed per update. DM and HDM require only a small number of scalar arithmetic operations. HECM additionally performs parameter lookup and observer matrix operations. DD evaluates recurrent matrix products and its MLP output head. At the nominal 1-Hz update rate, these medians occupy less than 0.043% of one sampling period. This deadline share describes the isolated estimator task and provides the execution margin available for integration with other BMS functions. The DD value in the cross-model comparison is the continuous-state deployment variant. Exact 2024-sample rolling-window semantics are evaluated separately because they recompute the full sequence at every sample.

For the cross-model embedded performance comparison, DD denotes the

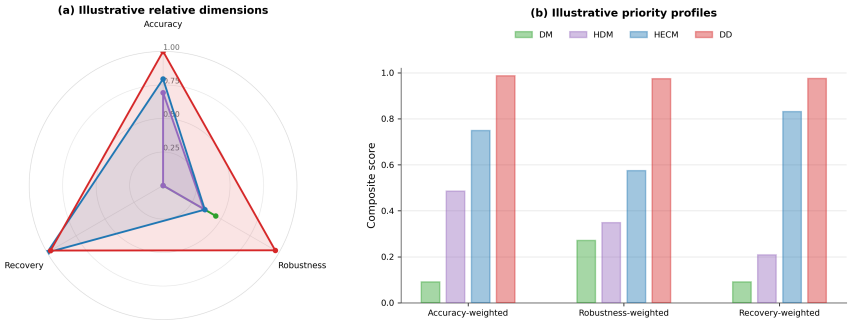

Figure 16: Decision-oriented synthesis for the four tested representatives. (a) Relative min–max-normalized accuracy, family-balanced robustness across eight evaluated disturbance families, and observed persistent paired recovery. Current gain and additive current offset use adverse directions from their matched signed sweeps. (b) Composite scores under three 60/20/20 priority profiles. The profiles are interpreted together with the scenario-level evidence in Figure 15.

continuous-state deployment variant used in the latency analysis. The compiled DM, HDM, HECM, and DD firmware occupies 27.0, 27.1, 470.1, and 115.9 KiB of flash, respectively. The corresponding measured peak runtime RAM is 2.4, 2.4, 2.5, and 4.1 KiB (Figure 19). These peak values combine static storage, dynamic allocation, and call-stack demand. HECM flash occupancy is dominated by its parameter surfaces, whereas the DD parameter storage accounts for most of its flash demand. The measurements describe the isolated SOC firmware with the shared SOH trace provided externally and leave resource margin for additional BMS tasks.

Figure 20 makes the DD deployment trade-off explicit, while Figure A.24 provides the corresponding latency distributions over all six hardware cell sequences. The same pruned weights are used in every mode. Exact rolling-window inference preserves the trained software semantics but requires roughly 724 ms median inference time, or 72.4% of the one-second period, and 67.3 KiB peak runtime RAM because the complete 2024-sample sequence is recomputed

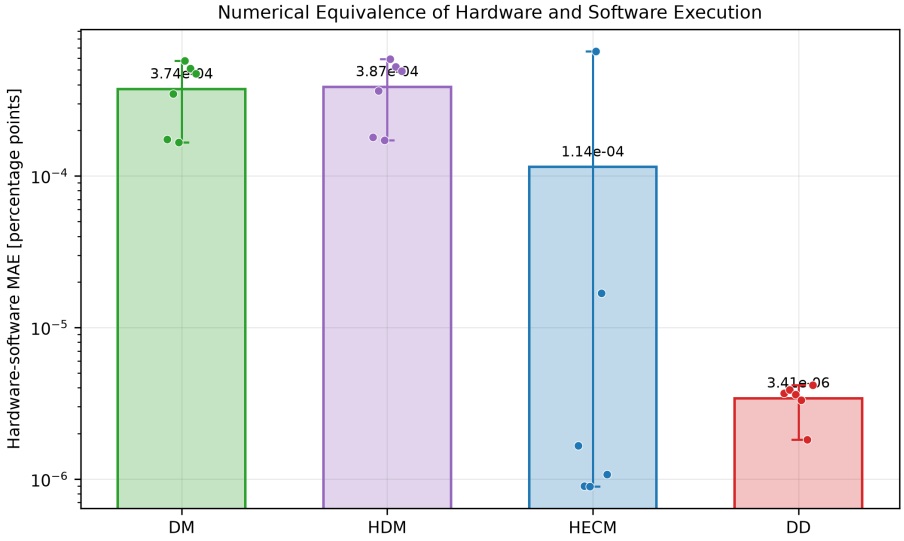

Figure 17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware–software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that all implementation differences are negligible relative to model error.

for every output. Continuous-state inference evaluates only the new recurrent step and reduces latency to approximately 0.425 ms and peak RAM to 4.1 KiB while retaining similar nominal error on the six sequences. Periodically resetting that state has a comparable 4.1 KiB peak but creates much larger transient errors. The mean of the cellwise maximum dataset-SOC errors is approximately 29 percentage points, and individual cells reach approximately 39 percentage points. Each reset removes the temporal context accumulated before that point, whereas continuous forwarding preserves it. Continuous-state execution therefore provides the practical hardware configuration. Its forwarded recurrent state defines a deployment mode that is evaluated separately from the primary rolling-window robustness benchmark.

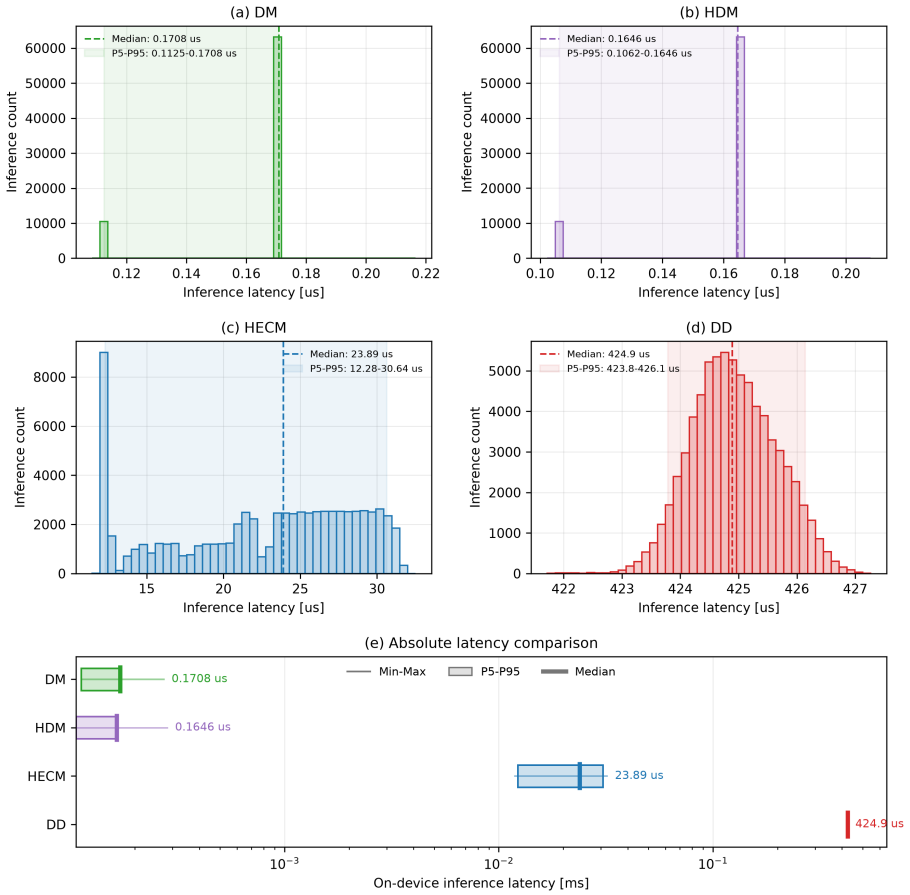

Figure 18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a–d) show per-inference distributions. Panel (e) compares minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model latency comparison.

5. Discussion

5.1. Structural interpretation across estimator classes

The four implementations expose distinct relationships between their update mechanisms and observed disturbance responses. DM has the largest nominal

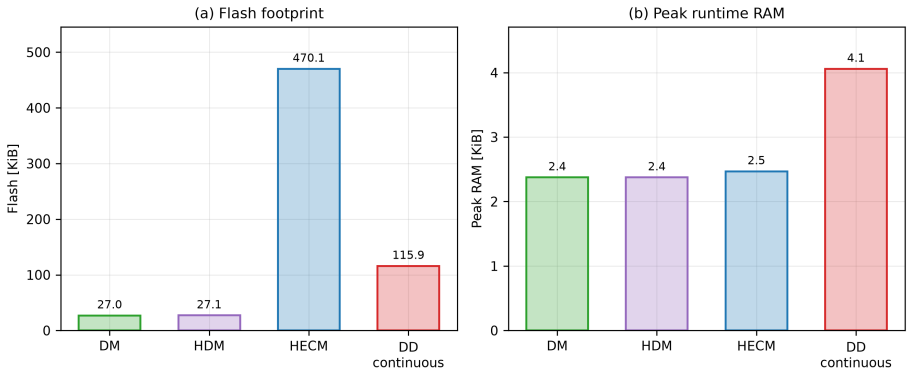

Figure 19: Embedded memory requirements of the isolated STM32 SOC implementations. (a) Compiled flash occupancy. (b) Peak runtime RAM measured during a representative holdout-test replay, combining statically allocated variable storage with maximum dynamic-memory allocation and call-stack use. DD denotes continuous-state execution, matching Figure 18. Exact rolling-window execution is compared separately in Figure 20.

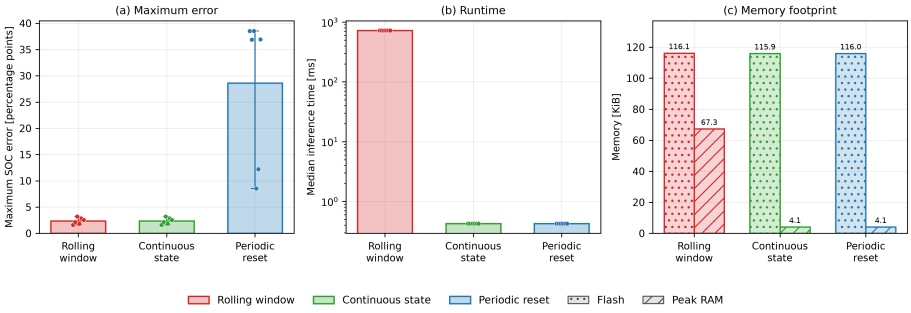

Figure 20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, compiled flash occupancy, and measured peak runtime RAM for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, not continuous forwarding, create the dominant accuracy loss.

error and reacts strongly to persistent current-gain and current-offset errors as well as missing samples because its current-integration structure lacks voltage-residual correction. Its practical full-charge anchor limits accumulated drift, while the lifecycle experiment shows that the reset restructures the gain-error contribution without removing it uniformly. HDM adapts the capacity denom-

inator through learned SOH and thereby reduces capacity mismatch, yet it retains the same charge-integration core. Its similar response to systematic current errors, missing charge information, and ADC quantization follows directly from this shared structure.

HECM combines current-driven state propagation with a voltage residual and has the strongest observed recovery profile under the tested initialization mapping. Additive current offset produces its largest global disturbance penalty. Among the remaining cases, high current noise and burst dropout are most consequential because corrupted or frozen inputs affect both state propagation and observer correction. The confidence interval for high current noise includes zero and shows substantial between-cell uncertainty. The supplementary lookup analysis in Appendix Appendix B shows that the largest local lookup–disturbance interaction is 0.00612 MAE, compared with a 0.1763 HECM penalty from the adverse current offset itself. The disturbance response is therefore governed primarily by the corrupted input in this case, although absolute accuracy and observer re-anchoring remain dependent on lookup calibration.

DD has the lowest nominal cell-macro error and the smallest adverse penalties under both systematic current-error tests. It is also comparatively resilient to isolated sample loss. Event-aligned voltage spikes expose a recurrent transient that full-window MAE hides. The selected voltage and derivative features and the rolling recurrent context provide a plausible transmission path for this response and identify a concrete target for feature ablation and disturbance-aware training.

The ground-truth SOH comparison demonstrates why the auxiliary state estimator must be considered part of the complete SOC pipeline. Part of the HDM and HECM baseline error is inherited through the estimated SOH input, whereas the DD response reflects a learned interaction between SOH and the remaining features. A more accurate auxiliary input therefore improves a physically defined capacity relation more directly than it improves a multivariate learned mapping. At the same time, the limited change in several paired disturbance penalties shows that the shared SOH estimator is not the dominant cause

of every robustness response. This distinction prevents errors of the auxiliary SOH service from being attributed incorrectly to the SOC core and supports evaluation of the complete causal pipeline as the primary configuration.

At class level, these representative results identify transferable mechanisms rather than a universal ranking. Direct-measurement methods provide transparent and deterministic updates with minimal computational and memory demand. Their update structure primarily propagates accumulated charge, however, and offers no continuous mechanism for reconstructing missing charge information or correcting systematic integration errors. Reliable current calibration, uninterrupted sampling, and physical anchoring are therefore central to their robustness. Adding learned SOH information can reduce capacity mismatch in a hybrid direct-measurement method, but it does not by itself add voltage feedback or temporal correction to the underlying integration core.

Model-based ECM observers extend this structure with a physically defined correction mechanism. The voltage residual allows HECM to re-anchor its internal SOC state and is consistent with its strong observed recovery behavior. This correction does not provide the same breadth of measurement history as a learned multichannel sequence representation. Its effectiveness remains constrained by the observability of the operating point, the flat OCV–SOC relation of LFP cells, the calibrated parameter surfaces, and disturbances that affect both current-driven state propagation and observer correction. The data-driven representative instead combines current, voltage, temperature, derived quantities, and their temporal development within one recurrent context. This broader context provides a plausible mechanism for its low nominal error and its comparatively small penalties under several systematic-error and sample-loss scenarios. The results therefore support the value of multivariate temporal information without establishing that every data-driven estimator will outperform every physical model.

Temporal context is not exclusively beneficial. The voltage-spike experiment shows that a localized disturbance can propagate through selected features and the recurrent state, while the periodic-reset experiment shows that removing ac-

cumulated context can create large transient errors. The same mechanism that allows DD to combine information across time can therefore transmit disturbances or become unavailable when its state is handled incorrectly. Structural pruning makes this context model smaller by reducing the recurrent width, but it does not resolve these temporal failure mechanisms. Robust data-driven deployment consequently requires representative training coverage, disturbance-aware evaluation, careful feature selection, and a state-handling strategy that matches the intended embedded execution mode.

5.2. Limitations of nominal accuracy

Nominal MAE describes average agreement with the dataset ground truth under clean inputs. The disturbance and recovery experiments add the DD voltage-spike transient, HECM re-anchoring behavior, shared-SOH feature coupling, and model-specific responses to random noise, systematic sensor errors, and missing information. Embedded measurements add latency, memory, and inference-state semantics. These dimensions create a deployment profile for each implementation, while the weighting analysis shows how application priorities reshape their combined scores. Estimator selection can therefore begin with the relevant fault and recovery requirements and then incorporate the available compute, memory, and state-service architecture.

5.3. Implications for BMS deployment

For BMS deployment, estimator classes should be selected according to the dominant sensing risks, recovery requirements, and hardware constraints. Direct-measurement methods are attractive when transparency, deterministic execution, and minimal memory are primary requirements, provided that current calibration, reliable sampling, and physical anchoring can be ensured. Hybrid methods can add adaptive health information or learned correction to an existing physical core. Their benefit and their remaining sensitivities depend on which part of that core is modified. Model-based ECM observers provide feedback correction and stronger re-anchoring potential at the cost of parameterization effort and larger storage requirements. Data-driven methods provide broad

flexibility for representing nonlinear dependencies and temporal context, but require representative training data and explicit validation of feature-mediated transients and recurrent-state behavior.

The STM32 results show that hardware resources do not constitute a fundamental deployment barrier for any of the four isolated SOC cores on the selected microcontroller. Every implementation fits into the available memory and completes its update within the one-second sampling interval. This result does not make computational constraints irrelevant, because the measured resource profiles differ substantially. DM and HDM provide the smallest and fastest implementations. HECM remains fast but requires 470.1 KiB of flash, which is dominated by its parameter surfaces. The continuous-state DD implementation requires more computation and runtime memory than the other representatives, yet its median latency of approximately 0.425 ms and peak runtime RAM of 4.1 KiB demonstrate that recurrent neural estimation is feasible within the tested embedded environment.

The DD implementation separates two complementary deployment decisions. Structured pruning reduces recurrent width before firmware export and yields a dense compact network that can be executed efficiently without specialized sparse kernels. Recurrent execution semantics then determine how often this fixed network is evaluated and how its temporal context is retained. Exact rolling-window execution requires approximately 724 ms and 67.3 KiB peak runtime RAM because it repeatedly evaluates the complete input sequence. Continuous-state forwarding evaluates only the new recurrent step and retains similar nominal error while using a small fraction of the sampling period. Periodic state resets preserve the low computational cost but produce large transient errors because they remove accumulated temporal information. The reduction from rolling-window to continuous-state cost is therefore an implementation-semantic gain rather than an additional compression result. Model architecture and execution strategy must be designed together, with the chosen state semantics validated against the required estimation behavior. The firmware measurements cover flash, maximum call-stack use, dynamic-memory demand,

and numerical equivalence. Integration of the shared SOH-LSTM, energy measurement, pack communication, and concurrent BMS scheduling forms the next system-level validation stage. Table 3 relates the measured estimator profiles to concrete deployment priorities.

Figure 16 complements the recommendation map by combining the measured accuracy, robustness, and recovery dimensions into relative priority profiles. The three profiles assign 60% to their named dimension and 20% to each remaining dimension. Family-balanced, equal-scenario, and high-severity aggregations produce different score distances and change the HDM–HECM order. This sensitivity turns the score into a configurable decision aid, while the scenario heatmap identifies the disturbance-specific strengths and weaknesses behind each profile.

5.4. Limitations and future extensions

The conclusions are grounded in four specified implementations and the six-cell holdout test set from one controlled LFP aging dataset. Alternative reset logic, ECM structures, neural architectures, features, and training data can produce different estimator profiles. The low- and middle-load strata contain multiple cells, whereas the high-load stratum contains one exploratory trajectory. One low-load test trajectory also provides no mid-life or aged window. Figure A.21 makes these coverage limits explicit. The reported confidence intervals therefore quantify the available between-cell variation within this design. Transfer to other chemistries, cell formats, pack configurations, thermal environments, and field duty cycles requires corresponding cell-disjoint validation. Nominal accuracy is referenced to one consistently reconstructed offline SOC ground truth based on Coulomb counting and voltage anchors.

The primary cross-model campaign concentrates on the BMS measurement chain, signal availability and timing, and deployment-accessible initialization states. The systematic current-error sweep covers matched signs at fixed gain and offset magnitudes, but not time-varying calibration drift or interacting sensor errors. The separate HECM lookup analysis crosses local resistance, time-

Table 3: Decision-oriented recommendation map derived from the benchmark results. The table identifies the strongest observed representative under one dominant deployment priority and the stated protocol boundaries.

Primary deployment priority and selected representative	Main supporting benchmark evidence	Main trade-off to consider
Lowest nominal SOC error: DD	Lowest six-cell macro MAE (0.0258) and RMSE (0.0300)	Rolling-window execution is the most computationally expensive implementation
Strongest observed recovery from initialization mismatch: HECM	Lowest observed persistent-recovery-or-censor time and substantially fewer first-entry relapses than DD	Recovery is compared only after all estimators provide valid output
Lowest adverse systematic current-error penalties: DD	Matched signed tests give ΔMAE 0.0038 at 3% gain error and 0.0401 at 50 mA offset, below the other representatives	Temperature-offset and local voltage-spike sensitivity remain visible
Smallest observed periodic/random sample-loss penalty: DD	ΔMAE 0.0011/0.0026	The duration-matched dropout effect is non-positive but highly variable across cells
Lowest deterministic compute cost: DM/HDM	Sub-microsecond median SOC-core latency and approximately 27 KiB flash	Integration structure is sensitive to gain error and missing charge information
Practical fast neural deployment: DD continuous state	Approximately 0.425 ms median and 4.1 KiB peak runtime RAM	Different recurrent semantics from the primary rolling-window benchmark

constant, and OCV deviations with the complete set of tested disturbance subcases and paired initialization recovery. These one-at-a-time perturbations do not define probability distributions for lookup error and do not cover combined parameter deviations. Broader parameter mismatch, pack-level interactions, and supervisory BMS faults form a separate validation layer. The initialization experiment maps an SOC-equivalent offset into each estimator's available state and compares recovery once all estimators provide valid output. The result-

ing times characterize operational re-anchoring under class-specific interventions rather than identical latent-state convergence. The continuous high-load hold-out replay complements the test-set windows with a mechanism-focused lifecycle view. The hourly SOH service represents gradual capacity fade, and channel-removal experiments would further isolate the DD voltage-spike transmission path. HECM uses HPPC-derived parameter surfaces identified from the model-development sets and held fixed throughout evaluation. The sensitivity analysis therefore characterizes local disturbance interactions around this calibrated implementation rather than arbitrary lookup-table error or every possible HECM design.

The hardware profile covers the isolated SOC firmware with SOH supplied as an external causal trace. Runtime RAM is measured during a representative holdout-test replay, including complete rolling-window DD inference. The next hardware stage combines the SOC and SOH services, measures energy demand, and evaluates estimator scheduling alongside communication and supervisory BMS tasks. Broader data campaigns can then extend the disturbance protocol to interacting faults, additional operating environments, and pack-level closed-loop behavior.

6. Conclusion

This study uses representative implementations from the direct-measurement, hybrid, model-based, and data-driven SOC-estimation families to establish a common robustness and embedded-performance benchmark. Each implementation was selected to expose a characteristic update or correction mechanism under the same causal protocol. The resulting evidence connects these mechanisms with the strengths and weaknesses that become relevant when an estimator is transferred from nominal software evaluation to practical BMS operation.

For direct-measurement methods, the benchmark confirms the advantages of transparent operation, deterministic execution, and very low resource demand. It also shows why reliable current sensing, uninterrupted charge information,

and physical anchoring are central to this family. Hybridization can improve an existing estimator by adding learned health information or adaptive correction, but its effect depends on the part of the underlying method that is modified. Capacity adaptation improves the treatment of ageing without automatically removing the disturbance sensitivities of a current-integration core. Model-based ECM observers add physically interpretable voltage-feedback correction and strong re-anchoring capability. These benefits are accompanied by calibration effort, parameter storage, and dependence on the validity of the physical model over the intended operating range.

The data-driven family shows particularly strong potential for future BMS applications. The recurrent representative combines the lowest nominal error with strong results across several disturbance families, demonstrating the value of learning nonlinear and temporal relationships from multichannel battery data. Its deployment path also shows how this capability can be made compatible with embedded execution. Structured pruning reduces the GRU width from 96 to 67 units before export, while continuous-state forwarding reduces median inference latency from approximately 724 ms for exact rolling-window recomputation to 0.425 ms and peak runtime RAM from 67.3 KiB to 4.1 KiB without a relevant loss of nominal accuracy on the evaluated sequences. Periodic reset does not provide the same behavior because removing the hidden context creates pronounced transients. Compact recurrent state estimation is therefore feasible on the selected microcontroller, but its practical quality depends on both network preparation and stateful execution. Representative training coverage, disturbance-aware evaluation, and careful handling of input features and recurrent state remain essential for exploiting this potential reliably.

The broader consequence for all estimator families is that nominal MAE alone cannot establish suitability for BMS deployment. Accuracy must be evaluated together with disturbance-specific robustness, recovery behavior, numerical equivalence, inference latency, flash occupancy, and runtime RAM. This multidimensional benchmark does more than compare four implementations. It reveals the mechanisms that determine class-specific behavior and provides a practical

basis for improving a selected approach through sensing design, physical-model adaptation, disturbance-aware neural training, or embedded optimization. The same framework can be extended to complete SOC–SOH execution, concurrent BMS tasks, additional operating environments, and pack-level applications.

Appendix A. Dataset Split, Holdout Coverage, and Evaluation Protocol

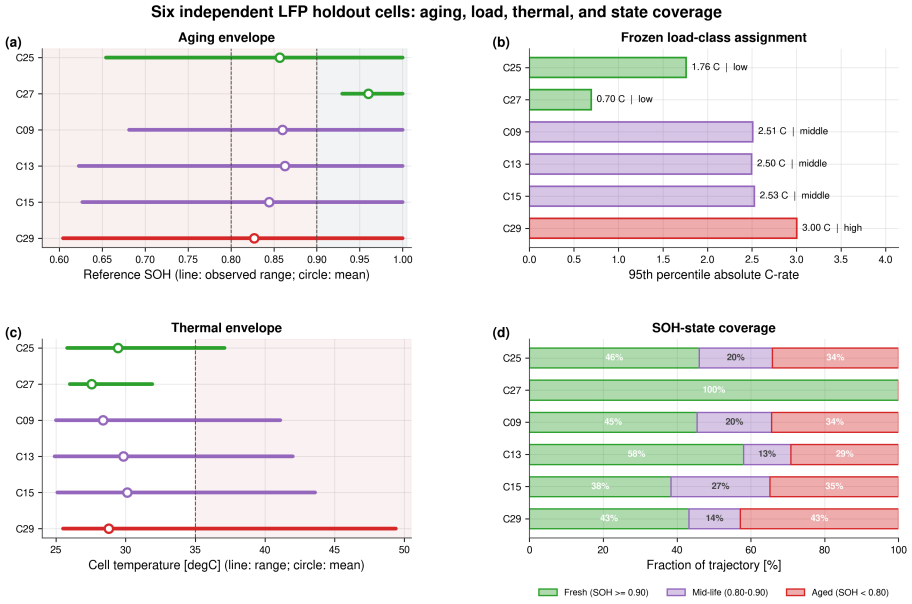

Figure A.21: Quantitative coverage of the six-cell independent holdout test set. Panels show observed SOH, the descriptive load-class assignment, thermal envelope, and full-trajectory SOH-state coverage. The high-load group contains one exploratory trajectory.

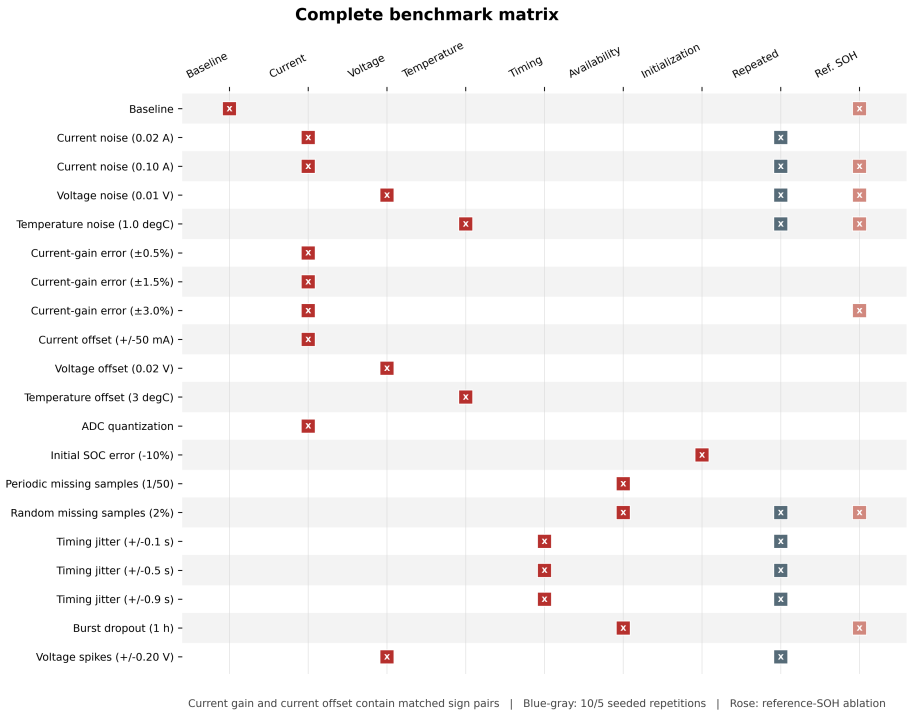

Figure A.22: Complete 20-scenario benchmark matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel. Blue-gray markers denote repeated stochastic cases. Rose markers identify paired reference-SOH ablations. Each current-gain magnitude and the additive current-offset case contain matched positive and negative sign sublevels.

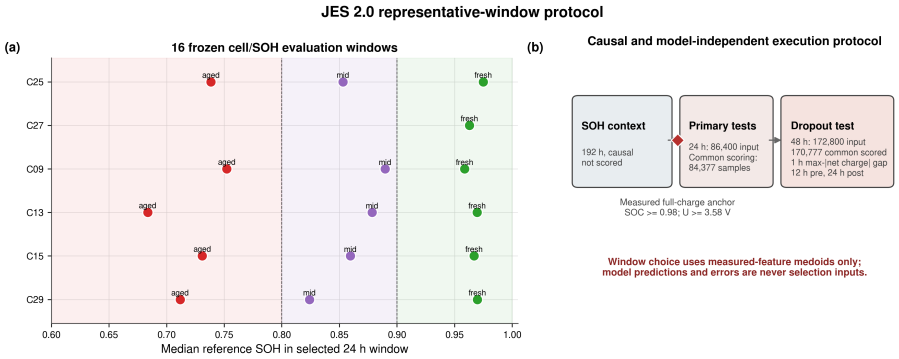

Figure A.23: Representative-window protocol. Sixteen cell/SOH windows are selected using measured-feature medoids. Every estimator is scored over the same interval once the rolling DD input context is complete. Standard windows cover 24 h after the causal SOH context, while the dropout case uses 48 h and places the one-hour gap at the eligible interval with maximum absolute net charge.

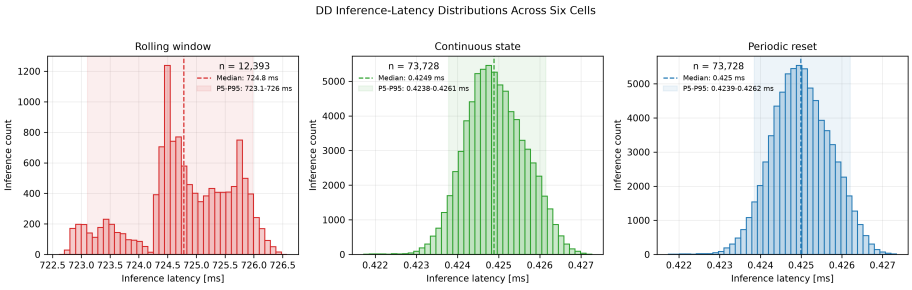

Figure A.24: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

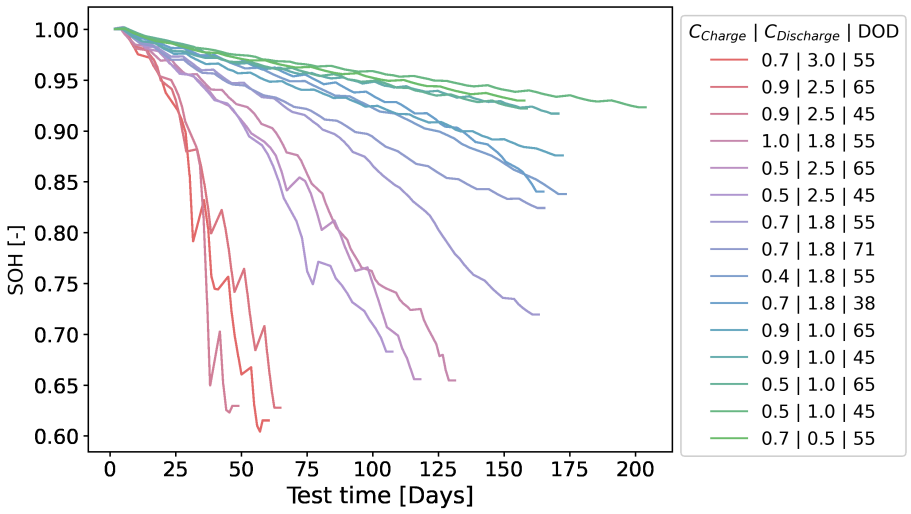

Figure A.25: Measured SOH trajectories across the complete LFP aging campaign. The campaign is divided into a six-cell training set, a three-cell validation set, and a six-cell independent holdout test set. The legend reports the applied charge C-rate, discharge C-rate, and depth of discharge for each trajectory. The plotted SOH values originate from repeated capacity checkups and are interpolated between checkup anchors. The exact cell-to-set assignment and measured coverage are reported in Table A.4.

Table A.4: Measured coverage and exact cell-to-set assignment for the fixed training, validation, and independent holdout test split. The 95th-percentile absolute C-rate is calculated over each complete 1-Hz trajectory. Load classes are defined only for the holdout test set. Its high-load class contains one cell and is exploratory.

Cell	Split	Duration [d]	SOH range	T range [°C]	P95 $\lvert C_{\mathrm{rate}} \rvert$	Test-set load
C01	Training	198.6	0.922–1.000	24.3–29.4	1.01	–
C03	Training	153.3	0.921–1.000	25.9–30.7	1.00	–
C05	Training	166.1	0.917–1.000	25.2–30.6	1.00	–
C11	Training	113.2	0.656–1.000	25.6–42.8	2.48	–
C17	Training	158.7	0.719–1.000	25.8–34.9	1.75	–
C23	Training	167.8	0.837–1.000	26.1–37.2	1.74	–
C07	Validation	167.6	0.874–1.000	25.2–32.0	1.01	–
C19	Validation	160.4	0.840–1.000	26.3–33.9	1.75	–
C21	Validation	160.8	0.824–1.000	26.2–37.6	1.75	–
C09	Test	102.6	0.682–1.000	25.0–41.1	2.51	Middle
C13	Test	44.7	0.623–1.000	24.9–42.0	2.50	Middle
C15	Test	60.7	0.627–1.000	25.1–43.6	2.53	Middle
C25	Test	127.1	0.655–1.000	25.8–37.1	1.76	Low
C27	Test	152.0	0.930–1.000	26.0–31.9	0.70	Low
C29	Test	56.0	0.604–1.000	25.5–49.4	3.00	High*

Appendix B. HECM Lookup-Table Sensitivity

The cross-model benchmark evaluates one fixed HECM implementation whose lookup surfaces were identified exclusively from the training and validation data and then held constant. This supplementary HECM-only analysis tests whether its disturbance response depends strongly on local lookup calibration changes. The R_i, R_1, and R_2 surfaces are jointly scaled by -10% or $+10\%$. The τ_1 and τ_2 surfaces are likewise jointly scaled by -10% or $+10\%$, and the OCV surfaces are shifted by $-10\,\mathrm{mV}$ or $+10\,\mathrm{mV}$. These six one-at-a-time perturbations and the nominal lookup are evaluated across 22 measurement and signal-integrity subcases plus the paired initialization intervention, all original seeds, and all 16 frozen windows. The additive current-offset extension uses the same $\pm 0.05\,\mathrm{A}$ inputs as the cross-model benchmark. The combined design comprises 9184 HECM executions.

Main-campaign disturbances use their lookup-matched main baseline, the one-hour dropout uses a lookup-matched 48-h baseline, and initialization recovery uses a separate paired lookup-matched baseline. For each disturbance, the lookup interaction is defined as the scenario ΔMAE under the perturbed lookup minus the corresponding ΔMAE under the nominal lookup. Results are first averaged within cells and then aggregated using the same equal-weight cell bootstrap as the primary benchmark. The analysis remains outside the cross-model robustness score because it probes one model's local parameter sensitivity rather than a shared input disturbance.

The local lookup changes shift baseline HECM MAE from 0.0314 to 0.0373, compared with 0.0337 for the nominal lookup. Across the 22 measurement and signal-integrity subcases, the largest absolute interaction is 0.006116 MAE. It occurs for the $-0.05\,\mathrm{A}$ current offset under the $+10\%$ time-constant scaling, and its cell-bootstrap interval includes zero. The interaction is small relative to the nominal-lookup adverse current-offset penalty of 0.1763 MAE. The largest current-gain interaction remains 0.000443 MAE. Several smaller interactions have intervals that exclude zero, which identifies systematic local changes with-

out making the lookup calibration the dominant source of the observed disturbance penalties.

Initialization recovery exposes a separate boundary. Five lookup perturbations retain mean persistent recovery-or-censor times between 1.18 and 1.41 h, close to the nominal 1.20 h. Scaling all resistance surfaces by −10% instead increases the equal-weight cell mean to 5.10 h because one low-load holdout cell remains outside the recovery band through the 24-h horizon. The other five cells do not become censored. The lookup analysis therefore supports local stability of most measurement-disturbance responses, but it does not establish lookup independence of absolute accuracy or observer re-anchoring.

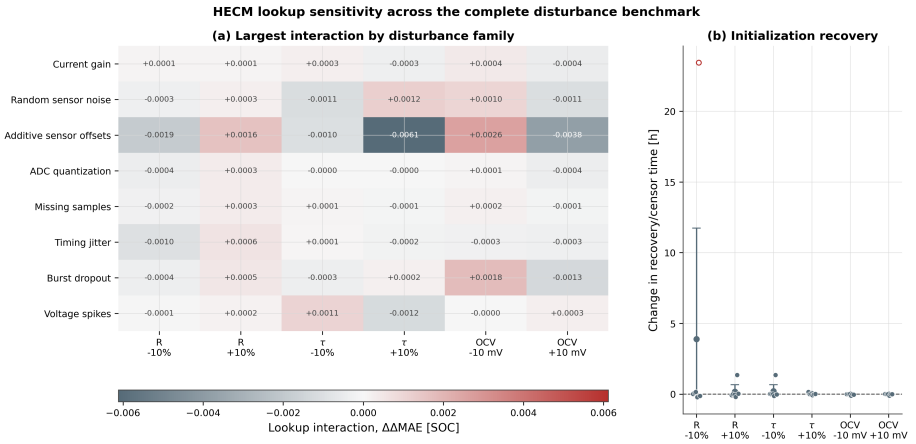

Figure B.26: HECM lookup sensitivity across the complete disturbance benchmark. Panel (a) reports the largest lookup interaction within each of the eight measurement and signal-integrity robustness families under local resistance, time-constant, and OCV perturbations. The additive-offset family includes current, voltage, and temperature offsets. Panel (b) reports the corresponding change in persistent initialization recovery-or-censor time. Points show equal-weight cell means. Open red points identify cell means containing a censored recovery window. Each run uses the appropriate lookup-matched main, dropout, or recovery baseline.

Table B.5: HECM lookup sensitivity across the complete disturbance benchmark. The largest robustness interaction is selected from 22 measurement and signal-integrity subcases after equal-weight cell aggregation. Bracketed counts report subcases whose 95% interaction interval excludes zero. Recovery is the persistent recovery-or-censor time from the paired initialization analysis, with the equal-weight censored-cell percentage in brackets.

Lookup condition	Baseline MAE	Max. robust $\lvert\Delta\Delta\text{MAE}\rvert$ [CI]	Source subcase	Recovery/censor [h] [censored]
Nominal lookup	0.0337	Reference [–]	–	1.20 [0%]
Resistance −10%	0.0351	0.00191 [6/22]	Current offset +50 mA	5.10 [17%]
Resistance +10%	0.0327	0.00162 [5/22]	Current offset +50 mA	1.38 [0%]
Time constants −10%	0.0343	0.00113 [6/22]	Current noise 0.10 A	1.41 [0%]
Time constants +10%	0.0335	0.00612 [5/22]	Current offset −50 mA	1.22 [0%]
OCV −10 mV	0.0314	0.00260 [6/22]	Current offset +50 mA	1.18 [0%]
OCV +10 mV	0.0373	0.00378 [5/22]	Current offset +50 mA	1.19 [0%]

Appendix C. Supplementary Numerical Results

Table C.6: Six-cell baseline cell-macro statistics with hierarchical 95% confidence intervals.

Estimator	MAE [SOC]	RMSE [SOC]	P95 error [SOC]
DM	0.0690 [0.0499, 0.0823]	0.0740 [0.0532, 0.0881]	0.1088 [0.0768, 0.1307]
HDM	0.0412 [0.0301, 0.0515]	0.0442 [0.0319, 0.0557]	0.0649 [0.0449, 0.0846]
HECM	0.0337 [0.0246, 0.0434]	0.0388 [0.0286, 0.0493]	0.0644 [0.0468, 0.0821]
DD	0.0258 [0.0189, 0.0325]	0.0300 [0.0223, 0.0366]	0.0508 [0.0375, 0.0619]

Table C.7: Exact paired baseline-MAE comparisons across the six-cell holdout test set. Differences are defined as estimator B minus estimator A. Holm correction is applied across the six model-pair tests.

Estimator A	Estimator B	Mean difference [SOC]	95% CI	d_z	p_{exact} / p_{Holm}
DM	HDM	-0.0278	[-0.0381, -0.0171]	-1.93	0.0313 / 0.1875
DM	HECM	-0.0352	[-0.0470, -0.0224]	-2.00	0.0313 / 0.1875
DM	DD	-0.0431	[-0.0558, -0.0286]	-2.29	0.0313 / 0.1875
HDM	HECM	-0.0075	[-0.0162, 0.0015]	-0.61	0.1563 / 0.1875
HDM	DD	-0.0154	[-0.0243, -0.0084]	-1.39	0.0313 / 0.1875
HECM	DD	-0.0079	[-0.0176, -0.0005]	-0.68	0.0938 / 0.1875

Table C.8: Selected cell-macro scenario penalties relative to paired baseline. Current-gain and additive current-offset values use adverse directions from their matched signed sweeps.

Scenario	DM	HDM	HECM	DD
Current noise, $\sigma_I = 0.10\,\text{A}$	-0.0009	+0.0000	+0.0038	-0.0000
Current gain, adverse 3%	+0.0108	+0.0101	+0.0060	+0.0038
Current offset, adverse 50 mA	+0.2467	+0.2727	+0.1763	+0.0401
Voltage offset, +0.02 V	+0.0000	+0.0005	-0.0040	+0.0038
Temperature offset, +3 °C	+0.0000	+0.0002	+0.0002	+0.0074
ADC quantization	+0.0038	+0.0035	-0.0018	+0.0002
Random missing samples, 2%	+0.0080	+0.0081	+0.0039	+0.0026
Timing jitter, ±0.9 s	-0.0003	+0.0003	+0.0006	+0.0003
Burst dropout, 1 h	+0.0050	+0.0209	+0.0239	-0.0093
Voltage spikes, ±0.20 V	+0.0000	-0.0003	+0.0002	-0.0001

Table C.9: Sensitivity of the illustrative normalized robustness score to the aggregation rule.

Estimator	Family balanced	Equal scenario	Highest level plus offsets
DM	0.452	0.549	0.573
HDM	0.351	0.476	0.469
HECM	0.360	0.552	0.562
DD	0.967	0.793	0.693

Table C.10: Canonical paired initialization recovery and target-hardware summary. Recovery values are equal-weight cell means over the common evaluation interval. Endpoints already satisfied when all estimators first provide valid output are recorded as conservative upper bounds.

Metric	DM	HDM	HECM	DD
First 300-s entry/censor time [h]	22.28	17.21	1.12	1.29
Persistent recovery/censor time [h]	22.28	22.28	1.20	2.60
Recovery excess-error AUC [SOC h]	1.735	1.738	0.078	0.116
First-entry censored fraction	0.89	0.67	0.00	0.00
Persistent-recovery censored fraction	0.89	0.89	0.00	0.00
Relapse fraction after first hold	0.00	0.22	0.17	0.78
Median SOC-core latency [μs]	0.171	0.165	23.9	424.9[a]
Flash [KiB]	27.0	27.1	470.1	115.9[a]
Peak runtime RAM [KiB]	2.4	2.4	2.5	4.1[a]

[a]Continuous-state DD. Exact rolling-window execution requires approximately 724 ms and 67.3 KiB peak RAM.

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Florian Rzepka: Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing—original draft preparation, writing—review and editing, visualization. Qinnan Chen: Writing—review and editing. Julia Kowal: Resources, supervision, project administration, funding acquisition, writing—review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT, DeepL, and Grammarly in order to improve the linguistic quality and style of the

manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Data availability

The dataset is publicly available through its cited TU Berlin repository record [28]. Code, model configurations, benchmark definitions, random seeds, result summaries, hardware timing records, and analysis procedures are maintained in a public GitHub repository. The repository contains the neural model assets and HECM parameterization used for the reported experiments.

References

[1] P. Shrivastava, P. A. Naidu, S. Sharma, B. K. Panigrahi, A. Garg, Review on technological advancement of lithium-ion battery states estimation methods for electric vehicle applications, Journal of Energy Storage 64 (2023) 107159. doi:10.1016/j.est.2023.107159.

[2] F. Guo, L. D. Couto, G. Mulder, K. Trad, G. Hu, O. Capron, K. Haghverdi, A systematic review of electrochemical model-based lithium-ion battery state estimation in battery management systems, Journal of Energy Storage 101 (2024) 113850. doi:10.1016/j.est.2024.113850.

[3] J. Yao, J. Kowal, Towards a smarter battery management system: A critical review on deep learning-based state of charge estimation of lithium-ion batteries, Energy and AI 21 (2025) 100585. doi:10.1016/j.egyai.2025.100585.

[4] X. Li, J. Jiang, C. Zhang, L. Y. Wang, L. Zheng, Robustness of SOC Estimation Algorithms for EV Lithium-Ion Batteries against Modeling Errors and Measurement Noise, Mathematical Problems in Engineering 2015 (2015) 1–14. doi:10.1155/2015/719490.

[5] F. Naseri, C. Barbu, A. Jensen, P. Setoodeh, J. R. Baena, N. Hevele, E. Schaltz, Toward Reliable Wireless BMS: Robust Battery State Estimation Under Noise and Uncertainty, IEEE Transactions on Vehicular Technology (2025) 1–12doi:10.1109/TVT.2025.3638613.

[6] F. Berger, D. Joest, E. Barbers, K. Quade, Z. Wu, D. U. Sauer, P. Dechent, Benchmarking battery management system algorithms - Requirements, scenarios and validation for automotive applications, eTransportation 22 (2024) 100355. doi:10.1016/j.etran.2024.100355.

[7] Z. A. Khan, P. Shrivastava, S. M. Amrr, S. Mekhilef, A. A. Algethami, M. Seyedmahmoudian, A. Stojcevski, A Comparative Study on Different Online State of Charge Estimation Algorithms for Lithium-Ion Batteries, Sustainability 14 (12) (2022) 7412. doi:10.3390/su14127412.

[8] Y. Gao, G. L. Plett, G. Fan, X. Zhang, Enhanced state-of-charge estimation of LiFePO4 batteries using an augmented physics-based model, Journal of Power Sources 544 (2022) 231889. doi:10.1016/j.jpowsour.2022.231889.

[9] Y. Gao, K. Liu, C. Zhu, X. Zhang, D. Zhang, Co-Estimation of State-of-Charge and State-of- Health for Lithium-Ion Batteries Using an Enhanced Electrochemical Model, IEEE Transactions on Industrial Electronics 69 (3) (2022) 2684–2696. doi:10.1109/TIE.2021.3066946.

[10] J. Shen, W. Ma, J. Xiong, X. Shu, Y. Zhang, Z. Chen, Y. Liu, Alternative combined co-estimation of state of charge and capacity for lithium-ion batteries in wide temperature scope, Energy 244 (2022) 123236. doi:10.1016/j.energy.2022.123236.

[11] C. Wang, N. Cui, Z. Cui, H. Yuan, C. Zhang, Fusion estimation of lithium-ion battery state of charge and state of health considering the effect of temperature, Journal of Energy Storage 53 (2022) 105075. doi:10.1016/j.est.2022.105075.

[12] P. Mondal, D. Bhavsar, K. Mittal, M. Mittal, Estimating State-of-Charge in Lithium-Ion Batteries Through Deep Learning Techniques: A Comparative Evaluation, IEEE Access 12 (2024) 78773–78786. doi:10.1109/ACCESS.2024.3408220.

[13] D. P. Pau, A. Aniballi, Tiny Machine Learning Battery State-of-Charge Estimation Hardware Accelerated, Applied Sciences 14 (14) (2024) 6240. doi:10.3390/app14146240.

[14] S. Giazitzis, M. Sakwa, S. Leva, E. Ogliari, S. Badha, F. Rosetti, A Case Study of a Tiny Machine Learning Application for Battery State-of-Charge Estimation, Electronics 13 (10) (2024) 1964. doi:10.3390/electronics13101964.

[15] Y. Mazzi, H. Ben Sassi, A. Gaga, F. Errahimi, State of charge estimation of an electric vehicle's battery using tiny neural network embedded on small microcontroller units, International Journal of Energy Research 46 (6) (2022) 8102–8119. doi:10.1002/er.7713.

[16] M. A. Kamali, A. C. Caliwag, W. Lim, Novel SOH Estimation of Lithium-Ion Batteries for Real-Time Embedded Applications, IEEE Embedded Systems Letters 13 (4) (2021) 206–209. doi:10.1109/LES.2021.3078443.

[17] B. Gou, Y. Xu, X. Feng, An Ensemble Learning-Based Data-Driven Method for Online State-of-Health Estimation of Lithium-Ion Batteries, IEEE Transactions on Transportation Electrification 7 (2) (2021) 422–436. doi:10.1109/TTE.2020.3029295.

[18] W. Li, N. Sengupta, P. Dechent, D. Howey, A. Annaswamy, D. U. Sauer, Online capacity estimation of lithium-ion batteries with deep long short-term memory networks, Journal of Power Sources 482 (2021) 228863. doi:10.1016/j.jpowsour.2020.228863.

[19] F. Rzepka, P. Hematty, M. Schmitz, J. Kowal, Neural Network Architecture

for Determining the Aging of Stationary Storage Systems in Smart Grids, Energies 16 (17) (2023) 6103. doi:10.3390/en16176103.

[20] BQ79616 16-Series Battery Monitor, Balancer, and Integrated Hardware Protector (2023).

[21] Design Considerations of Current Sensing With BQ769x2 Family for Battery Management System (2022).

[22] TMCS1108 3% Functional Isolation Hall-Effect Current Sensor With ±100-V Working Voltage (2024).

[23] TLE4972 high precision coreless current sensor for automotive applications (2023).

[24] O. Aiello, Electromagnetic Susceptibility of Battery Management Systems' ICs for Electric Vehicles: Experimental Study, Electronics 9 (3) (2020) 510. doi:10.3390/electronics9030510.

[25] M. Liu, D. Geißler, S. Bian, B. Zhou, P. Lukowicz, Assessing the Impact of Sampling Irregularity in Time Series Data: Human Activity Recognition As A Case Study (Jan. 2025). arXiv:2501.15330, doi:10.48550/arXiv.2501.15330.

[26] 10s battery pack monitoring, balancing, and comprehensive protection, 50-A discharge reference design (2024).

[27] A. Aqachtoul, K. Karam, A. Elamrani, Y. Alidrissi, M. Najib, A. Rochd, M. Bakhouya, MGFARM: An IoT and Edge-Driven Framework for Smart and Sustainable Agricultural Practices, in: Y. Mejdoub, A. Elamri, M. Kardouchi (Eds.), Connected Objects, Artificial Intelligence, Telecommunications and Electronics Engineering, Vol. 1584, Springer Nature Switzerland, Cham, 2026, pp. 419–425. doi:10.1007/978-3-032-01536-5_64.

[28] F. Rzepka, Design of Experiments for Cyclic Aging of LFP 18650 Cells: Impact of Charge/Discharge C-Rates, Depth of Discharge, and Temperature on Battery Degradation (Nov. 2025). doi:10.14279/DEPOSITONCE-24800.

[29] K. Siebertz, D. Van Bebber, T. Hochkirchen, Statistische Versuchsplanung, Springer Berlin Heidelberg, Berlin, Heidelberg, 2017. doi:10.1007/978-3-662-55743-3.